

Fine-Tuning a Language Model on MHQA Dataset

Generative AI — Assignment 2

Bhukya Jhansi	160123748007
Bommareddy Sri Sanjana	160123748008
Budama Lakshmi Pragna Manasvi	160123748009

Project Guide: **Panigrahi Srikanth**
Assistant Professor, Department of AI&ML

CBIT, Hyderabad

Outline

- 1 Problem Statement
- 2 Dataset
- 3 Methodology
- 4 Model Architecture
- 5 Results
- 6 Deployment
- 7 Conclusion

Problem Statement

Objective

Fine-tune a pre-trained Language Model on the **MHQA (Mental Health Question Answering)** dataset and deploy it as a web application.

- Mental health is a growing global challenge
- Lack of standard benchmarking datasets for mental health QA
- LLMs have shown great potential in healthcare applications
- MHQA provides expert-verified multiple-choice QA pairs

MHQA Dataset

- Source: PubMed abstracts
- 2,475 gold standard instances
- ~56.1k pseudo-labeled pairs
- Expert-verified QA pairs

MHQA-b Dataset

- Extended version
- Includes validity labels
- Filtered to valid questions only

Combined Statistics

Property	Value
Total rows	58,617
After filtering	~52,000
Train split	90%
Val split	10%
Answer choices	4

Dataset — Domains & Question Types

Mental Health Domains

- **Anxiety** disorders
- **Depression**
- **Trauma** / PTSD
- **Obsessive/Compulsive** issues

Question Types

- **Factoid** — factual recall
- **Diagnostic** — identifying conditions
- **Prognostic** — predicting outcomes
- **Preventive** — prevention strategies

Sample Question

"Which therapy is most effective for generalized anxiety disorder?"

(A) CBT (B) ECT (C) Hypnotherapy **(D) Recreational therapy**

① Data Preprocessing

- Combined MHQA + MHQA-b datasets
- Filtered valid questions, converted labels to 0-indexed
- 90/10 train/validation split

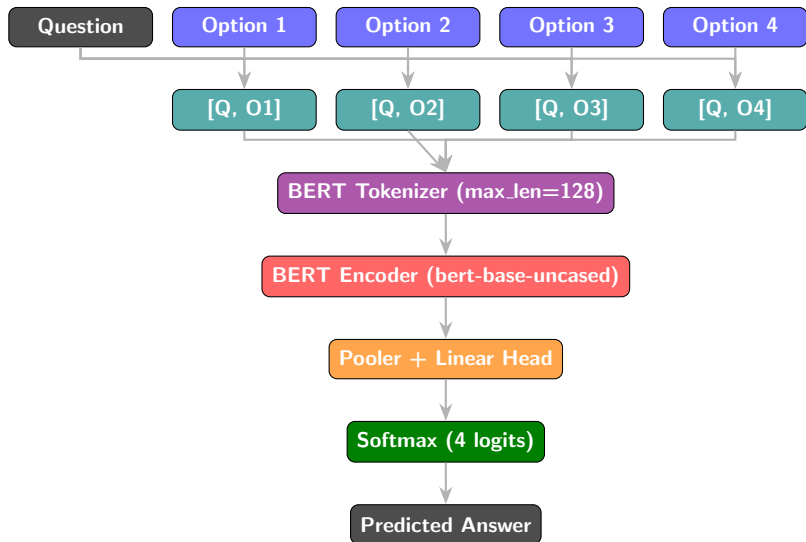
② Model Selection

- Base model: `bert-base-uncased`
- Added `BertForMultipleChoice` head
- 4-way classification per question

③ Training Configuration

- Epochs: 3, Batch size: 8
- Max sequence length: 128 tokens
- Mixed precision (FP16) on T4 GPU

Model Architecture



Results

Validation Performance

Metric	Score
Accuracy	79.31%
Val Loss	0.9244
Epochs	3

Unseen Data Testing

- Anxiety: ✓ Correct
- Depression: ✓ Correct
- OCD: ✓ Correct

Key Achievement

Model achieved **79.31% accuracy**, significantly above the random baseline of 25% for 4-choice QA.

Deployment

Tech Stack

- **Training:** Google Colab (T4 GPU)
- **Model Hosting:** Hugging Face Hub
- **UI Framework:** Gradio
- **Deployment:** Hugging Face Spaces
- **Code:** GitHub

Live Links

- **App:** HF Spaces — Live Demo
- **Model:** HF Hub — Model
- **Code:** GitHub Repository

Application UI

Features of the Web Application

- Enter any mental health question with 4 options
- Model predicts the correct answer instantly
- Shows confidence score for each option with visual bar
- Load example questions by topic (Anxiety/Depression/Trauma/OCD)
- Publicly accessible — no login required

<https://huggingface.co/spaces/bsrisanjana/mhqa-mental-health-qa>

Conclusion

- Successfully fine-tuned **BERT** on the MHQA dataset
- Achieved **79.31% accuracy** (vs 25% random baseline)
- Deployed a fully functional web app with public URL
- Model handles all 4 mental health domains effectively
- Demonstrates practical application of LLMs in healthcare QA

Future Work

- Try larger models (BioBERT, PubMedBERT)
- Expand to more mental health domains
- Add explanation generation for answers

Thank You!

Try the Live App!

Bhukya Jhansi • Bommareddy Sri Sanjana • Budama Lakshmi Pragna Manasvi
Guide: Panigrahi Srikanth • CBIT, Hyderabad